

Serverless Image/Video Processing

Services: Amazon S3, AWS Lambda, AWS Step Functions, Amazon Rekognition, Amazon SNS

Goal:

Automatically process images or videos uploaded to S3 using a serverless workflow. The system performs automated image analysis such as label detection using Amazon Rekognition and sends notifications through SNS.

Skills:

- Event-driven architecture
- Serverless orchestration
- Lambda automation
- AI-powered image analysis
- S3 event triggers
- Notification systems

Definition:

The Serverless Image/Video Processing System is designed to automatically process images or videos uploaded to an S3 bucket using fully serverless AWS services. The project leverages AWS Lambda for automation, Amazon Rekognition for AI-based image analysis, AWS Step Functions for workflow orchestration, and SNS for notifications. Whenever a user uploads an image or video to the S3 bucket, an event is triggered automatically. Lambda processes the uploaded file, invokes Rekognition for label detection or analysis, and sends the processing results through SNS notifications. This architecture eliminates manual intervention and server management by using event-driven serverless services.

Scope of the Project:

The scope of this project is to design and implement a fully automated serverless image and video processing pipeline using AWS cloud services.

This project covers:

- Automated Media Upload Processing:

Automatically detects new image or video uploads in the S3 bucket.

- **AI-Based Image Analysis:**

Uses Amazon Rekognition to identify objects, labels, scenes, or text within uploaded images.

- **Event-Driven Workflow Automation:**

Triggers Lambda functions automatically whenever new files are uploaded.

- **Notification System:**

Sends automated notifications containing processing results using SNS.

- **Workflow Orchestration:**

Uses Step Functions to coordinate multiple processing steps in sequence.

- **Fully Serverless Architecture:**

No server provisioning or infrastructure management is required.

Methodology:

1. Lambda + S3 Event Trigger + Rekognition – Best Method

- Uses all required AWS services: S3, Lambda, Rekognition, SNS, and Step Functions.
- Fully event-driven and serverless architecture.
- Automatically processes uploaded images without manual intervention.

- Supports scalable image analysis workflows.
- Cost-effective and Free-tier friendly for small workloads.
- Easy to maintain because no servers are managed manually.
- Suitable for modern cloud-native applications and AI-based automation.

2. EC2 + Python/OpenCV – Alternative Method

- Uses EC2 instances with custom Python scripts for image processing.
- Allows deep customization using OpenCV or ML libraries.
- Requires continuous server management and maintenance.
- Higher operational cost due to always-running EC2 instances.
- Requires OS patching, monitoring, and scaling management.
- Better suited for advanced custom video-processing workloads.

AWS Services used in the Project:

1. Amazon S3 (Simple Storage Service)

- **Definition:** Amazon S3 is an object storage service that provides scalable, secure, and durable cloud storage for files and media objects.
- **Purpose:** Stores uploaded images and videos.
- **Role:** Acts as the entry point of the workflow. Uploading a file triggers the automated processing pipeline.

2. AWS Lambda

- **Definition:** AWS Lambda is a serverless compute service that allows users to run code without provisioning or managing servers.
- **Purpose:** Processes uploaded files automatically.

Role:

- * Receives S3 upload events.
- * Invoke Step Function
- * Invokes Rekognition APIs.
- * Processes analysis results.
- * Sends notifications using SNS.

3. Amazon Rekognition

- **Definition:** Amazon Rekognition is an AI-powered image and video analysis service that uses machine learning to identify objects, scenes, labels, faces, and text.
- **Purpose:** Performs AI-based image analysis.
- **Role:** Detects labels and objects within uploaded images automatically.

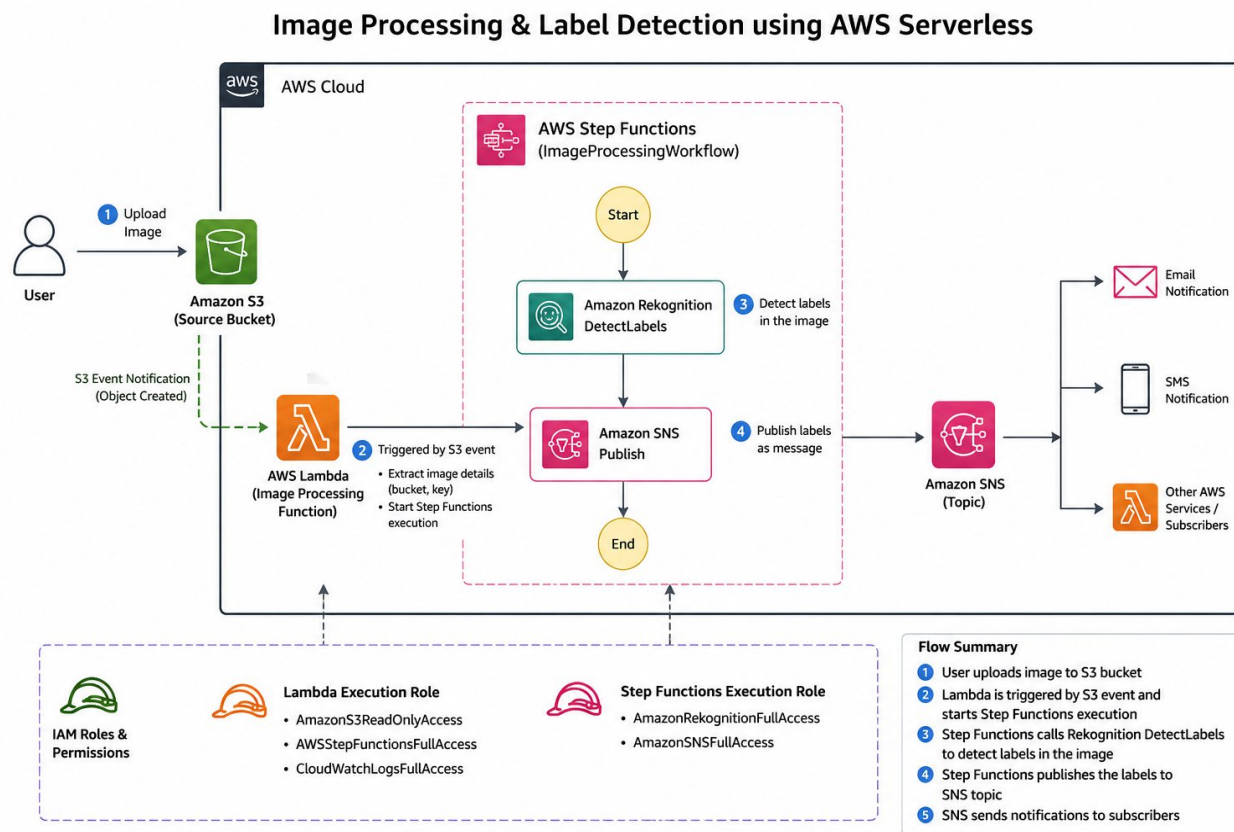
4. AWS Step Functions

- **Definition:** AWS Step Functions is a serverless orchestration service that coordinates multiple AWS services into automated workflows.
- **Purpose:** Controls workflow execution and sequencing.
- **Role:** Manages multi-step processing pipelines such as image analysis, resizing, and notifications.

5. Amazon SNS (Simple Notification Service)

- **Definition:** Amazon SNS is a fully managed messaging and notification service for sending alerts and notifications.
- **Purpose:** Sends processing notifications to users.
- **Role:** Sends email notifications containing image-processing results.

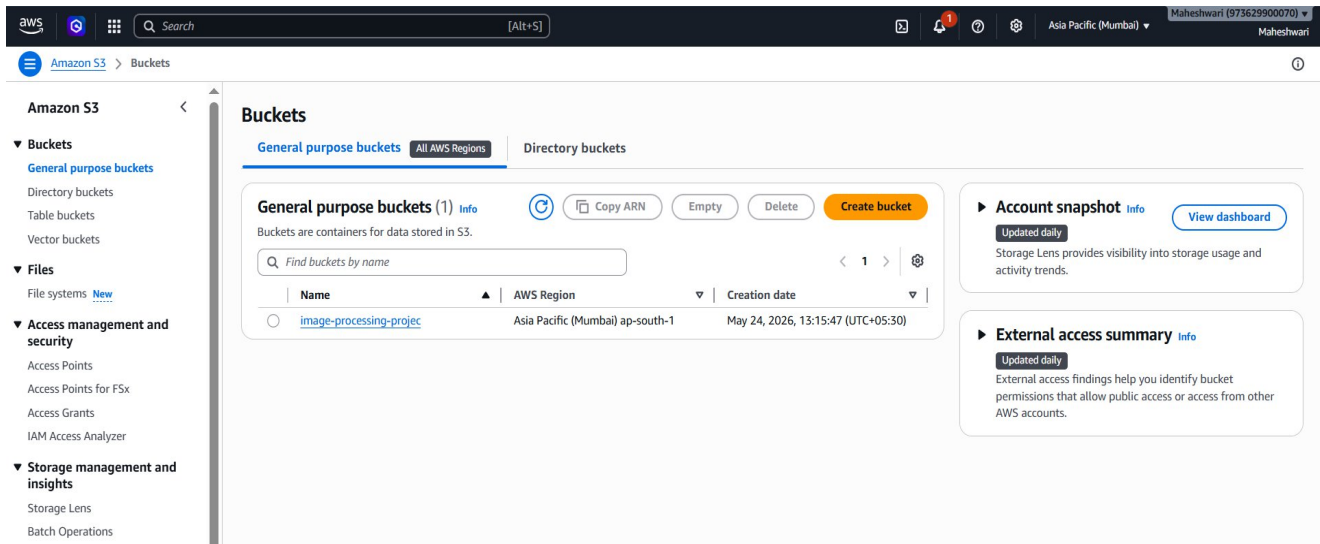
Architecture Diagram:



Implementation Steps to Do the Project:

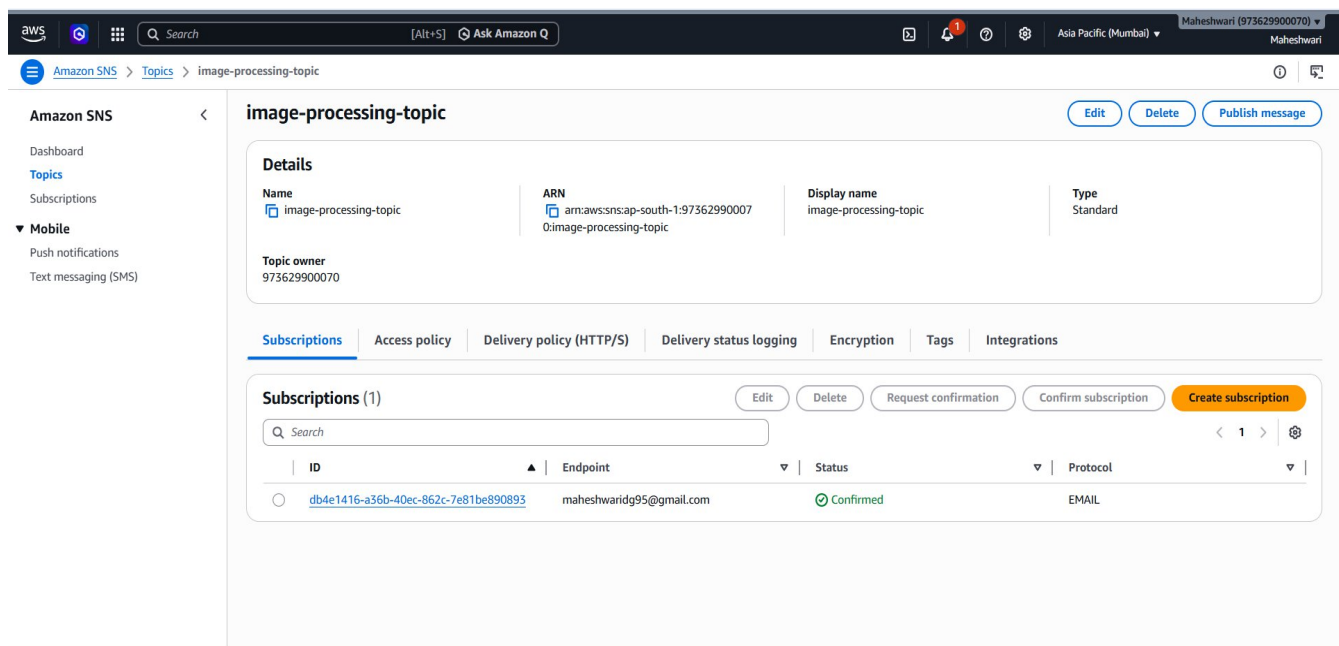
Step 1: Create S3 Bucket

- Go to AWS Console → S3 → Create Bucket.
- Enter Bucket Name: image-processing-project
- Select Region.
- Keep Block Public Access Enabled.
- Create Bucket.



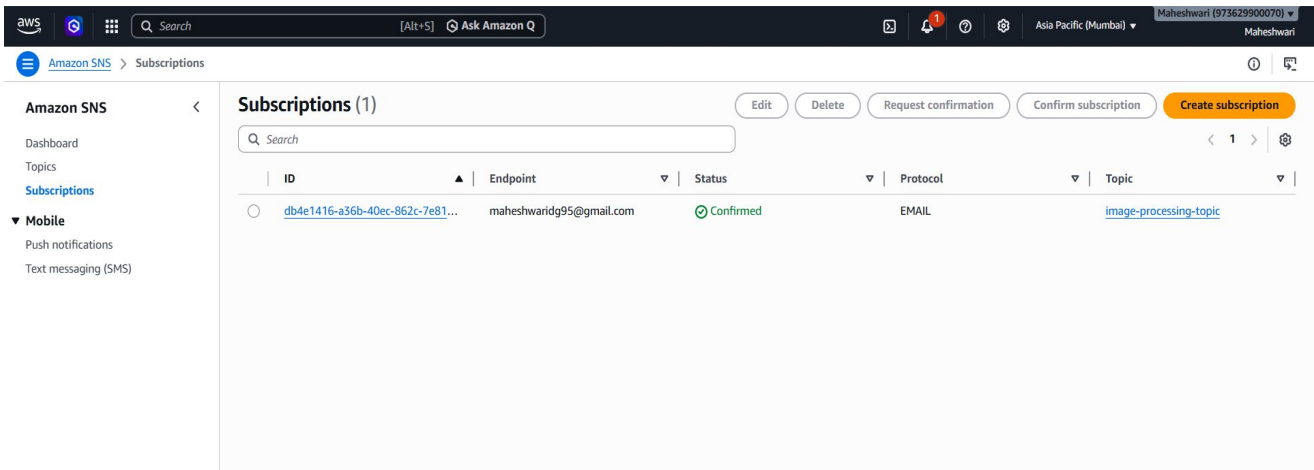
Step 2: Create SNS Topic

- Go to AWS Console → SNS → Topics.
- Click Create Topic.
- Type: Standard.
- Topic Name: image-processing-topic
- Create Topic.



Step 3: Create SNS Subscription

- Open the created SNS Topic.
- Click Create Subscription.
- Protocol: Email.
- Endpoint: Enter your email address.
- Create Subscription.
- Confirm subscription from your email inbox.

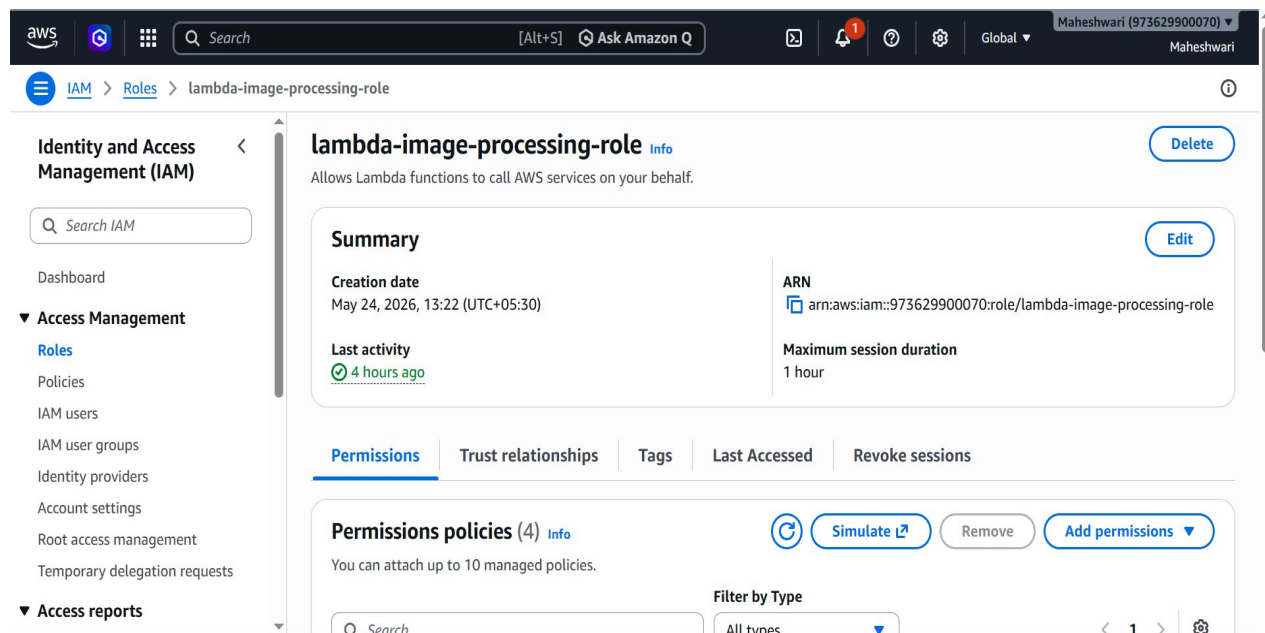


Step 4: Create IAM Role for Lambda

- Go to IAM → Roles → Create Role.
- Trusted Entity: Lambda.

Attach Policies:

- AWSLambdaBasicExecutionRole
- AmazonRekognitionFullAccess
- AmazonSNSFullAccess
- AmazonS3ReadOnlyAccess
- Role Name: lambda-image-processing-role



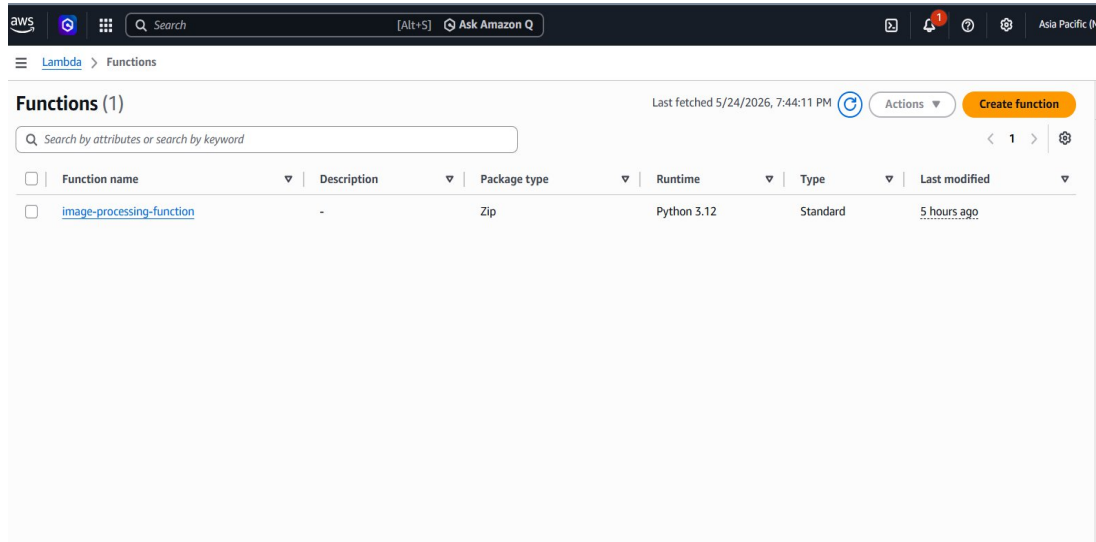
Step 5: Create Lambda Function

- Go to Lambda → Create Function.
- Author from Scratch.

Configuration:

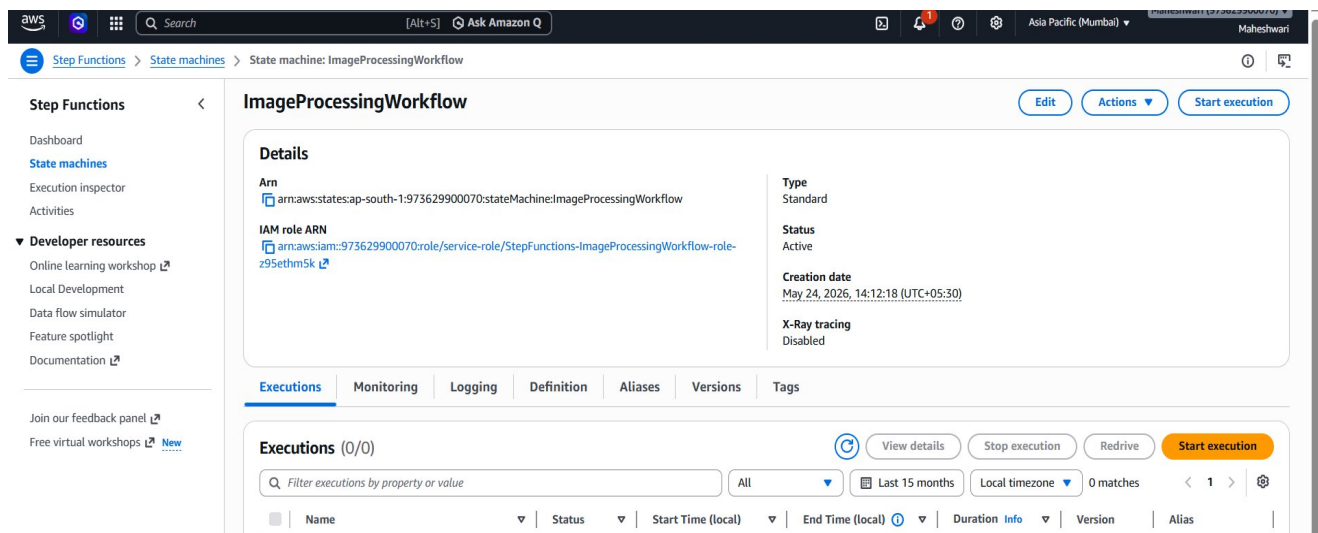
- Function Name: image-processing-function
- Runtime: Python 3.12

- Permissions: Use Existing Role
- Select Role: LambdaImageProcessingRole



Step 6: Create Step Functions Workflow

- Open Step Functions Console.
- Create State Machine.
- Add Lambda execution steps.
- Configure workflow sequence for:
 - * Image validation
 - * Rekognition analysis
 - * Notifications

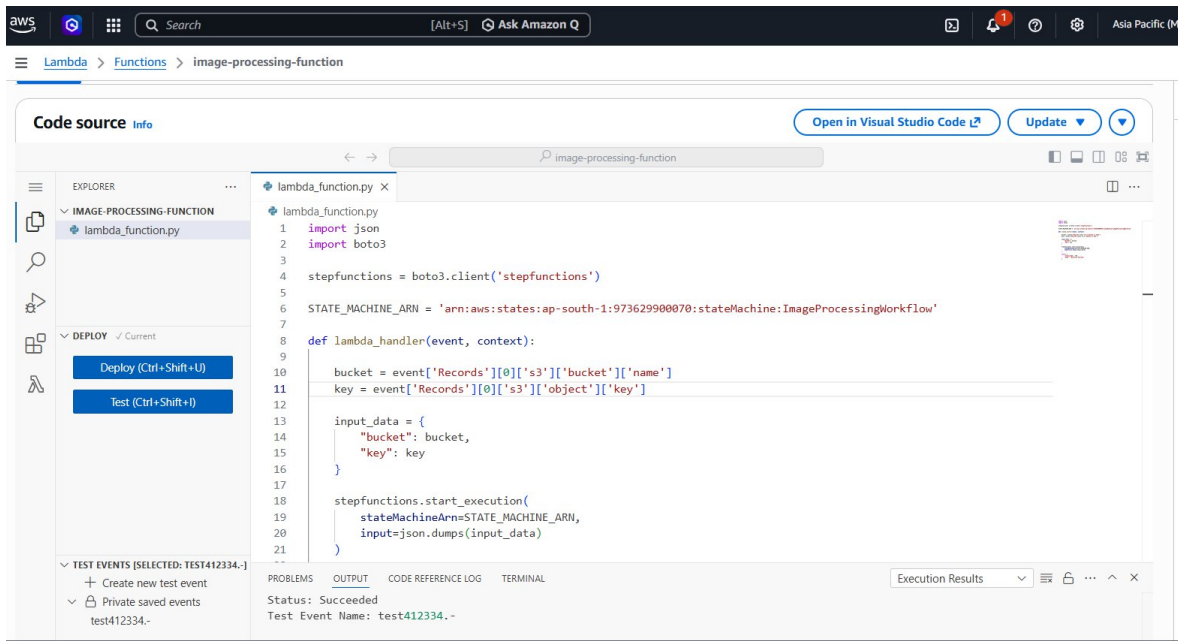


Step 6: Add Lambda Function Code

- Add Python code for Step Function which contains Rekognition label detection and SNS notifications.

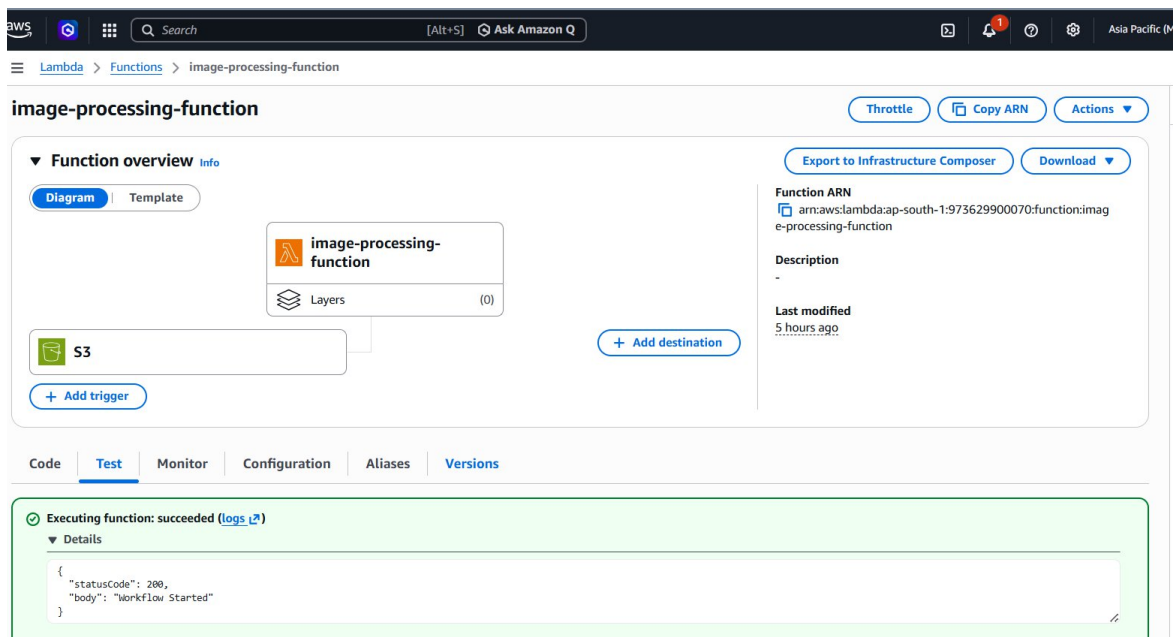
Step 7: Configure Step Function ARN

- Open Step Function.
- Copy Step Function ARN.
- Replace Step Function ARN inside Lambda code.



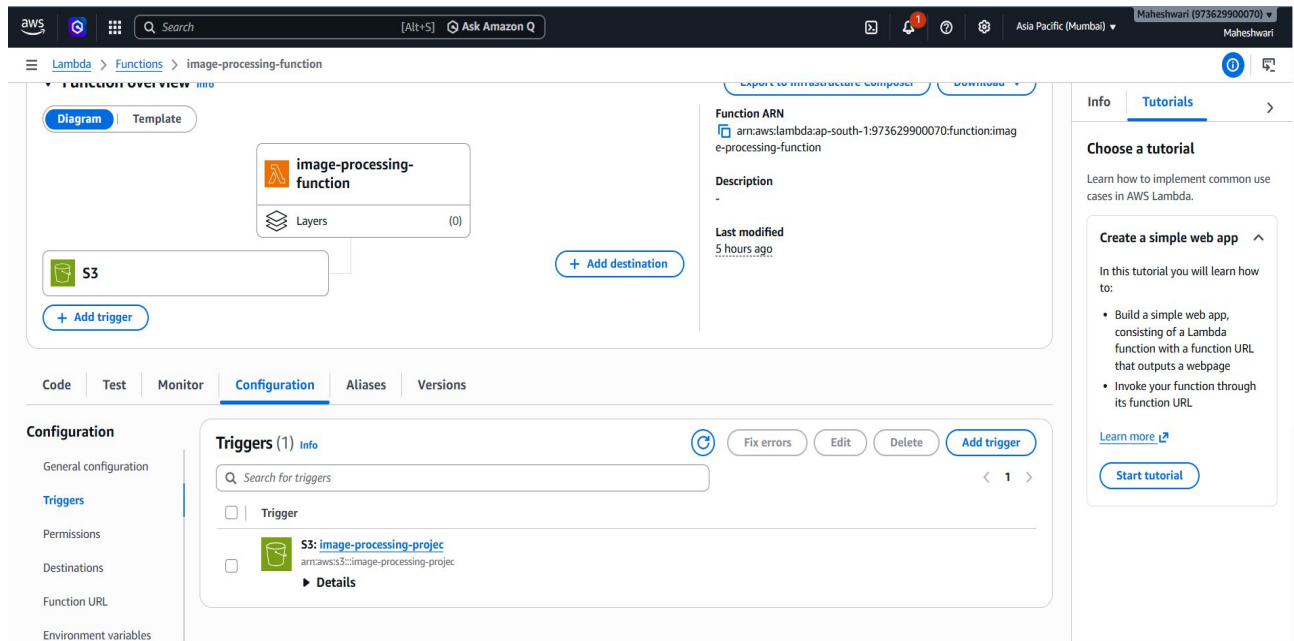
Step 8: Deploy and Test Lambda

- Click Deploy.
- Create Test Event.
- Run Test.
- Verify successful execution.



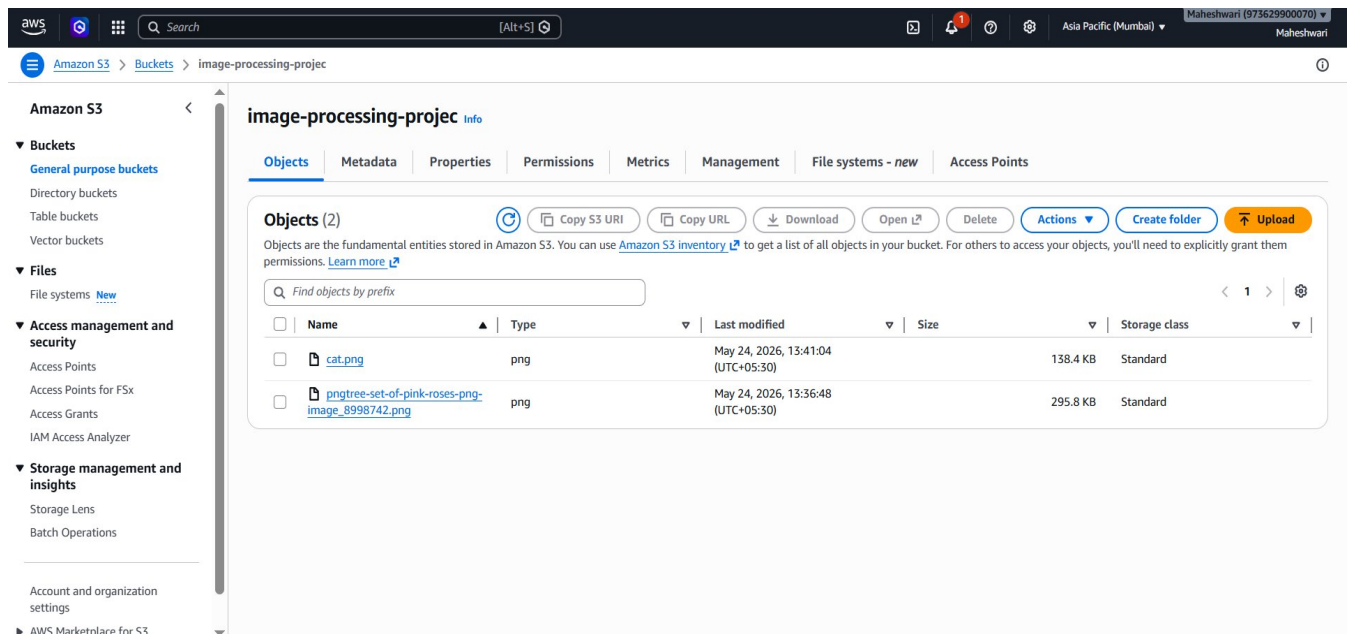
Step 9: Add S3 Trigger

- Inside Lambda, click Add Trigger.
- Select S3.
- Select Bucket: image-processing-projec
- Event Type: PUT
- Enable Trigger.



Step 10: Upload Test Image

- Open S3 Bucket.
- Upload sample image like Cat.jpg and Rose.png

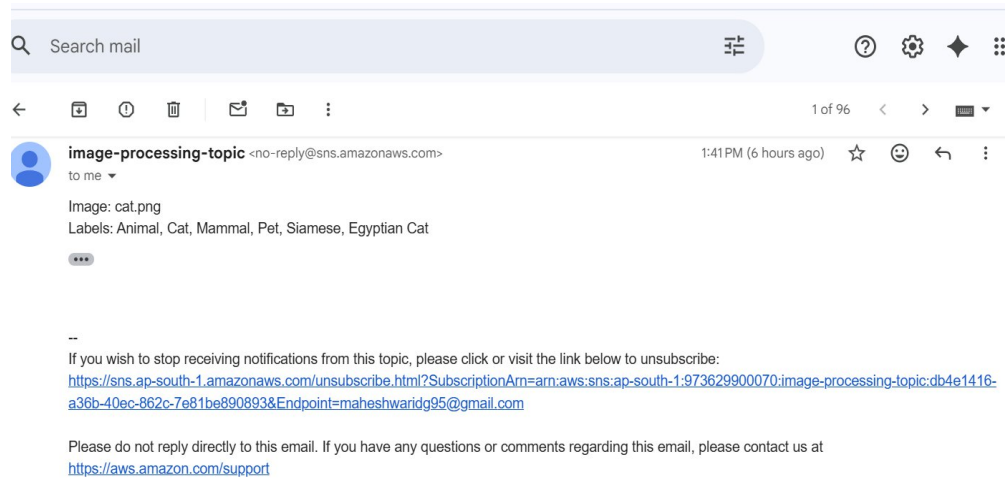
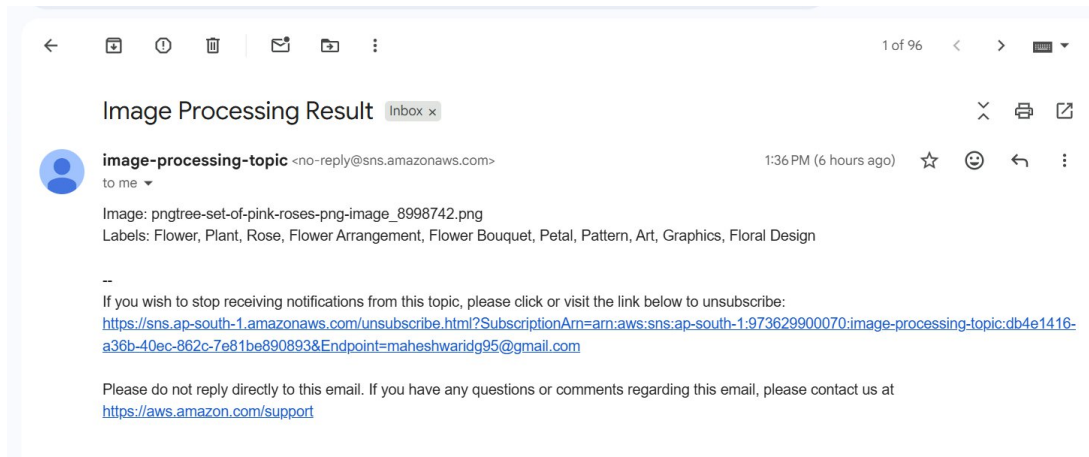


Step 11: Verify Email Notification

You should receive an email containing:

Image: dog.jpg

Labels: Dog, Animal, Pet, Mammal



Troubleshooting:

Issue 1: Lambda Timeout Error

Solution:

- Go to Lambda → Configuration.
- Click Edit.
- Set Timeout to 30 seconds.
- Save changes.

Issue 2: Access Denied Error

Solution:

- Attach required IAM policies:
- AmazonRekognitionFullAccess
- AmazonSNSFullAccess
- AmazonS3ReadOnlyAccess

Issue 3: SNS Email Not Received

Solution:

- Confirm SNS subscription from email inbox.

Issue 4: S3 Trigger Not Working

Solution:

- Reconfigure Lambda trigger.
- Verify bucket selection and PUT event configuration.

Conclusion:

The Serverless Image/Video Processing System provides a scalable, automated, and fully serverless solution for media processing in AWS. Using S3 event triggers, Lambda automation, Rekognition AI analysis, Step Functions orchestration, and SNS notifications, the system processes uploaded media efficiently without manual intervention. The architecture is highly scalable, cost-effective, and Free-tier friendly for small workloads. It demonstrates modern cloud-native design principles and showcases practical implementation of event-driven serverless workflows using AWS services.